

AI Persona — Evals Report

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95% Groundedness rate	5% Hallucination rate	0.685 Retrieval precision	0.962 Retrieval recall	17.29s Chat TTFB (deployed)
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1 · VOICE QUALITY

First-response latency. Component benchmark not run in this pass; end-to-end first-response latency measured by stopwatch on live calls (see voice_test_log.md). Architecture targets <2s via streaming STT, preemptive generation, and Flash-tier TTS (~75ms).

Transcription accuracy. Deepgram Nova-3 streaming STT; word-accuracy spot-checked against LiveKit session transcripts on live calls (logged in voice_test_log.md).

Task completion (booking). Booking verified end-to-end against the real Cal.com calendar (chat path confirmed a live booking; voice path uses the same booking tool).

2 · CHAT GROUNDEDNESS

Golden set of 20 cases run end-to-end against the **deployed** backend, graded by an LLM judge (gemini-2.5-flash) instructed to flag only contradictions/ fabrications (extra true detail allowed). Retrieval precision/recall computed from the expected-source labels per question over a 120-chunk corpus.

- **Groundedness:** 95% · **Hallucination:** 5%
- **Retrieval P/R:** 0.685 / 0.962
- Non-grounded cases flagged: book-1.

Case type	n	Grounded
adversarial	4	4/4
booking	1	0/1
factual	12	12/12
unknown	3	3/3

3 · THREE FAILURE MODES → ROOT CAUSE → FIX

- **Booking "time in the past".** The LLM resolved "tomorrow" against its training prior with no current date → built a stale ISO timestamp. *Fix:* inject current IST datetime into the prompt + force `start_iso` to be copied verbatim from the availability tool. Re-tested: booking confirmed on Cal.com.
- **Empty answers on Gemini (HTTP 400 mid-stream).** Gemini's OpenAI-compat endpoint streams parallel tool-calls with `index=None` ; an index-keyed accumulator collapsed them into unparseable args. *Fix:* key the accumulator by call id when index is absent; guard empty-query embeds. Re-tested: multi-search answers work.
- **Mid-session quota exhaustion (free tier).** Single key/model dependency 429'd. *Fix:* round-robin key rotation (cycles from last-working key, wraps) × model fallback (flash → flash-lite → 2.0-flash), skipping any failing key (429/403/5xx). Re-tested: serving continues after a key's daily bucket is exhausted.

4 · A CONSCIOUS TRADEOFF

In-memory vector store over a managed vector DB. The corpus is only 120 chunks (one résumé + ~20 repos), so embeddings are precomputed into a JSON file and loaded into a NumPy matrix at boot: retrieval is a single cosine matvec (<5ms), zero infra, zero per-query cost, and the voice agent does RAG *in-process* — removing an HTTP hop from the latency budget. The cost: refreshing the corpus needs a redeploy and there's no horizontal-scaling story for retrieval — both acceptable for a single-person corpus that changes at most weekly. (Related: a Gemini-only stack on the free tier traded a little latency, via key/model fallback, for ~\$0 running cost — see repo cost table.)

5 · WITH TWO MORE WEEKS

- **Voice:** clone Eeshu's real voice (ElevenLabs); speculative RAG prefetch from interim transcripts; warm-pool the Cloud agent to kill cold-start on first call.
- **RAG:** hybrid BM25 + dense retrieval with a reranker; nightly GitHub re-ingestion; per-claim inline citations rendered in chat.
- **Evals:** CI-gated eval run on every prompt change; an automated voice-eval rig (TTS-generated caller + scripted dialogues over SIP) to replace manual call logs.